

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 10 • Trigonometric Equations

With Answers

Questions with full worked solutions

- 49 pages
- Past papers up to January 2026

PURE 2 | TOPIC 10

Trigonometric Equations

Your classified route

20

QUESTIONS

173

MARKS

4

FAMILIES

5

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 10 | FAMILY 01

Trig Equations

WHAT THIS FAMILY ASKS

Recognise the trig equations demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**9 classified questions**

79 total marks

T10

10_CAST

10_CAST_CORE

CHAPTER
Trigonometric Equations

FAMILY
Trig equations

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12 marks

WMA12/01 OCTOBER 2019

Question 9

Past-paper question

WMA12-01_2019_Oct_Q09

9. Solutions based entirely on graphical or numerical methods are not acceptable in this question.

- (i) Solve, for $0 \leq \theta < 180^\circ$, the equation

$$3 \sin(2\theta - 10^\circ) = 1$$

giving your answers to one decimal place.

(4)

- (ii) The first three terms of an arithmetic sequence are

$$\sin \alpha, \quad \frac{1}{\tan \alpha} \quad \text{and} \quad 2 \sin \alpha$$

where α is a constant.

- (a) Show that $2 \cos \alpha = 3 \sin^2 \alpha$

(3)

Given that $\pi < \alpha < 2\pi$,

- (b) find, showing all working, the value of α to 3 decimal places.

(5)

T10 Trigonometric Equations 10_CAST Trig equations

10_CAST_CORE core family
move

WORKED SOLUTION

Question 9

WMA12/01 October 2019

WMA12-01_2019_Oct_Q09

1

Solve the sine equationLet $u = 2\theta - 10^\circ$. Then $\sin u = \frac{1}{3}$.

2

Find the valid angle valuesSince $0 \leq \theta < 180^\circ$, $-10^\circ \leq u < 350^\circ$. In this interval, $u = 19.471 \dots^\circ$ or $u = 160.529 \dots^\circ$.

3

Return to theta $\theta = \frac{u+10^\circ}{2}$, giving $\theta = 14.7^\circ$ and 85.3° to one decimal place.

4

Use the arithmetic sequence conditionFor three consecutive terms of an arithmetic sequence, second minus first equals third minus second. Hence $\frac{1}{\tan \alpha} - \sin \alpha = 2 \sin \alpha - \frac{1}{\tan \alpha}$.

5

Prove the required identityThis gives $\frac{2}{\tan \alpha} = 3 \sin \alpha$. Since $\frac{1}{\tan \alpha} = \frac{\cos \alpha}{\sin \alpha}$, we get $\frac{2 \cos \alpha}{\sin \alpha} = 3 \sin \alpha$, so $2 \cos \alpha = 3 \sin^2 \alpha$.

6

Form a quadratic in cos alphaUsing $\sin^2 \alpha = 1 - \cos^2 \alpha$, $2 \cos \alpha = 3(1 - \cos^2 \alpha)$, so $3 \cos^2 \alpha + 2 \cos \alpha - 3 = 0$.

7

Solve in the given interval $\cos \alpha = \frac{-1+\sqrt{10}}{3}$; the other root is less than -1 . Since $\pi < \alpha < 2\pi$ and cosine is positive, α is in the fourth quadrant.

8

State alpha $\alpha = 2\pi - \cos^{-1}\left(\frac{-1+\sqrt{10}}{3}\right) = 5.517$ radians to 3 decimal places.**FINAL ANSWER**

WMA12-01_2019_Oct_Q09

(i) $\theta = 14.7^\circ, 85.3^\circ$. (ii) (a) **Proven.** (ii) (b) $\alpha = 5.517$ radians.

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8 marks

WMA12/01 JANUARY 2021

Question 6

Past-paper question

WMA12-01_2021_Jan_Q06

6. (a) Show that the equation

$$\frac{3 \sin \theta \cos \theta}{2 \sin \theta - 1} = 5 \tan \theta \quad \sin \theta \neq \frac{1}{2}$$

can be written in the form

$$3 \sin^3 \theta + 10 \sin^2 \theta - 8 \sin \theta = 0$$

(4)

- (b) Hence solve, for $-\frac{\pi}{4} < x < \frac{\pi}{4}$

$$\frac{3 \sin 2x \cos 2x}{2 \sin 2x - 1} = 5 \tan 2x$$

giving your answers to 3 decimal places where appropriate.

(4)

T10 Trigonometric Equations 10_CAST Trig equations

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WORKED SOLUTION

Question 6

WMA12/01 January 2021

WMA12-01_2021_Jan_Q06

1

Clear the fractionStarting with $\frac{3 \sin \theta \cos \theta}{2 \sin \theta - 1} = 5 \tan \theta$, multiply by $(2 \sin \theta - 1) \cos \theta$.

2

Use tan thetaThis gives $3 \sin \theta \cos^2 \theta = 5 \sin \theta (2 \sin \theta - 1)$.

3

Use cos squaredSubstitute $\cos^2 \theta = 1 - \sin^2 \theta$: $3 \sin \theta (1 - \sin^2 \theta) = 10 \sin^2 \theta - 5 \sin \theta$.

4

Collect termsRearranging gives $3 \sin^3 \theta + 10 \sin^2 \theta - 8 \sin \theta = 0$.

5

Factor the sine equation $3s^3 + 10s^2 - 8s = 0$ gives $s(3s - 2)(s + 4) = 0$, where $s = \sin 2x$.

6

Use the valid sine valuesSince $-\frac{\pi}{2} < 2x < \frac{\pi}{2}$, the valid values are $\sin 2x = 0$ and $\sin 2x = \frac{2}{3}$.

7

Find x $2x = 0$ gives $x = 0$. Also $2x = \sin^{-1}\left(\frac{2}{3}\right)$, so $x = 0.364863... = 0.365$ to 3 decimal places.**FINAL ANSWER**

WMA12-01_2021_Jan_Q06

(a) Proven. (b) $x = 0, 0.365$.

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8 marks

WMA12/01 JANUARY 2022

Question 7

Past-paper question

WMA12-01_2022_Jan_Q07

7. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $-90^\circ < x < 90^\circ$, the equation

$$3 \sin(2x - 15^\circ) = \cos(2x - 15^\circ)$$

giving your answers to one decimal place.

(4)

- (ii) Solve, for $0 < \theta < 2\pi$, the equation

$$4 \sin^2 \theta + 8 \cos \theta = 3$$

giving your answers to 3 significant figures.

(4)

T10 Trigonometric Equations 10_CAST Trig equations

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WORKED SOLUTION

Question 7

WMA12/01 January 2022

WMA12-01_2022_Jan_Q07

1

Solve the first tangent equationLet $u = 2x - 15^\circ$. Then $3 \sin u = \cos u$, so $\tan u = \frac{1}{3}$.

2

Use the interval for u Since $-90^\circ < x < 90^\circ$, $-195^\circ < u < 165^\circ$.

3

Find x $u = 18.434 \dots^\circ$ or $u = -161.565 \dots^\circ$, so $x = 16.7^\circ$ or $x = -73.3^\circ$.

4

Rewrite the second equation $4 \sin^2 \theta + 8 \cos \theta = 3$.

5

Use sin squaredSubstitute $\sin^2 \theta = 1 - \cos^2 \theta$ to get $4(1 - \cos^2 \theta) + 8 \cos \theta = 3$.

6

Solve for cos thetaThis gives $4 \cos^2 \theta - 8 \cos \theta - 1 = 0$, so $\cos \theta = 1 \pm \frac{\sqrt{5}}{2}$.

7

Choose the valid cosineOnly $\cos \theta = 1 - \frac{\sqrt{5}}{2}$ lies between -1 and 1 .

8

State thetaFor $0 < \theta < 2\pi$, $\theta = 1.69$ or 4.59 to 3 significant figures.**FINAL ANSWER**

WMA12-01_2022_Jan_Q07

(i) $x = -73.3^\circ, 16.7^\circ$. (ii) $\theta = 1.69, 4.59$.

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6 marks

WMA12/01 MAY/JUNE 2022

Question 5

Past-paper question

WMA12-01_2022_MayJune_Q05

5. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Solve, for $-180^\circ < \theta \leq 180^\circ$, the equation

$$3 \tan(\theta + 43^\circ) = 2 \cos(\theta + 43^\circ)$$

(6)

T10 Trigonometric Equations 10_CAST Trig equations

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WORKED SOLUTION

Question 5

WMA12/01 May/June 2022

WMA12-01_2022_MayJune_Q05

1

Substitute u Let $u = \theta + 43^\circ$. The equation becomes $3 \tan u = 2 \cos u$.

2

Clear the cosine denominator $3 \frac{\sin u}{\cos u} = 2 \cos u$, so $3 \sin u = 2 \cos^2 u$.

3

Use cos squared $3 \sin u = 2(1 - \sin^2 u)$, so $2 \sin^2 u + 3 \sin u - 2 = 0$.

4

Factor the quadratic $(2 \sin u - 1)(\sin u + 2) = 0$.

5

Use the valid sine value $\sin u = -2$ is impossible, so $\sin u = \frac{1}{2}$.

6

Find u in the intervalSince $-180^\circ < \theta \leq 180^\circ$, $-137^\circ < u \leq 223^\circ$. Thus $u = 30^\circ$ or 150° .

7

Return to theta $\theta = u - 43^\circ$, so $\theta = -13^\circ$ or 107° .**FINAL ANSWER** $\theta = -13^\circ, 107^\circ$.

WMA12-01_2022_MayJune_Q05

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7 marks

WMA12/01 OCTOBER 2023

Question 3

Past-paper question

WMA12-01_2023_Oct_Q03

3. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Solve, for $0 < \theta \leq 360^\circ$ the equation

$$2 \tan \theta + 3 \sin \theta = 0$$

giving your answers, as appropriate, to one decimal place.

(5)

(b) Hence, or otherwise, find the smallest positive solution of

$$2 \tan(2x + 40^\circ) + 3 \sin(2x + 40^\circ) = 0$$

giving your answer to one decimal place.

(2)

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WORKED SOLUTION

Question 3

WMA12/01 October 2023

WMA12-01_2023_Oct_Q03

1

Factor the equation $2 \tan \theta + 3 \sin \theta = 0$ becomes $\frac{2 \sin \theta}{\cos \theta} + 3 \sin \theta = 0$.

2

Multiply by cos thetaSince $\tan \theta$ is defined, $\cos \theta \neq 0$. Multiplying by $\cos \theta$ gives $2 \sin \theta + 3 \sin \theta \cos \theta = 0$.

3

Factor $\sin \theta(2 + 3 \cos \theta) = 0$.

4

Solve the sine caseFor $0 < \theta \leq 360^\circ$, $\sin \theta = 0$ gives $\theta = 180^\circ$ or 360° .

5

Solve the cosine case $2 + 3 \cos \theta = 0$ gives $\cos \theta = -\frac{2}{3}$, so $\theta = 131.8^\circ$ or 228.2° to one decimal place.

6

Apply to the new angleFor $2 \tan(2x + 40^\circ) + 3 \sin(2x + 40^\circ) = 0$, the angle $2x + 40^\circ$ must be one of the solution angles, modulo 360° .

7

Find the smallest positive xThe smallest positive value comes from $2x + 40^\circ = 131.8 \dots^\circ$, so $x = \frac{131.8 \dots - 40}{2} = 45.9^\circ$ to one decimal place.**FINAL ANSWER**

WMA12-01_2023_Oct_Q03

(a) $\theta = 131.8^\circ, 180^\circ, 228.2^\circ, 360^\circ$. (b) $x = 45.9^\circ$.

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14 marks

WMA12/01 JANUARY 2024

Question 9

Past-paper question

WMA12-01_2024_Jan_Q09

9. In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 \leq x < 360^\circ$, the equation

$$\sin x \tan x = 5$$

giving your answers to one decimal place.

(6)

- (ii)

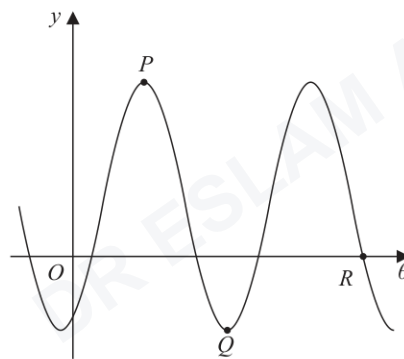


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = A \sin\left(2\theta - \frac{3\pi}{8}\right) + 2$$

where A is a constant and θ is measured in radians.

The points P , Q and R lie on the curve and are shown in Figure 1.

Given that the y coordinate of P is 7

- (a) state the value of A ,

(1)

- (b) find the exact coordinates of Q ,

(3)

- (c) find the value of θ at R , giving your answer to 3 significant figures.

(4)

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WORKED SOLUTION

Question 9

WMA12/01 January 2024

WMA12-01_2024_Jan_Q09

1

Rewrite the equation $\sin x \tan x = 5$ becomes $\frac{\sin^2 x}{\cos x} = 5$.

2

Use sine squaredMultiplying by $\cos x$ gives $\sin^2 x = 5 \cos x$. Since $\sin^2 x = 1 - \cos^2 x$, $1 - \cos^2 x = 5 \cos x$.

3

Solve for $\cos x$ $\cos^2 x + 5 \cos x - 1 = 0$, so $\cos x = \frac{-5 \pm \sqrt{29}}{2}$.

4

Choose the possible cosine value $\frac{-5 - \sqrt{29}}{2} < -1$, so the valid value is $\cos x = \frac{-5 + \sqrt{29}}{2}$.

5

Find x For $0 \leq x < 360^\circ$, this gives $x = 78.9^\circ$ or 281.1° to one decimal place.

6

Find A from P The point P is a maximum point on the sine curve. Its y coordinate is 7, so $A + 2 = 7$ and $A = 5$.

7

Locate Q Point Q is the next minimum after P , so $\sin\left(2\theta - \frac{3\pi}{8}\right) = -1$.

8

Find θ at Q For this minimum, $2\theta - \frac{3\pi}{8} = \frac{3\pi}{2}$, so $2\theta = \frac{15\pi}{8}$ and $\theta = \frac{15\pi}{16}$.

9

Find y at Q At the minimum, $y = 2 - 5 = -3$, so $Q = \left(\frac{15\pi}{16}, -3\right)$.

10

Set up R At R , $y = 0$, so $5 \sin\left(2\theta - \frac{3\pi}{8}\right) + 2 = 0$.

11

Choose the shown crossing

Thus $\sin\left(2\theta - \frac{3\pi}{8}\right) = -\frac{2}{5}$. The crossing R shown after the second maximum has $2\theta - \frac{3\pi}{8} = 3\pi + \sin^{-1}\left(\frac{2}{5}\right)$.

12

Find theta at R

$\theta = \frac{1}{2}\left(3\pi + \sin^{-1}\left(\frac{2}{5}\right) + \frac{3\pi}{8}\right) = 5.507\dots$, so $\theta = 5.51$ to 3 significant figures.

FINAL ANSWER

WMA12-01_2024_Jan_Q09

(i) $x = 78.9^\circ, 281.1^\circ$. (ii)(a) $A = 5$. (ii)(b) $Q = \left(\frac{15\pi}{16}, -3\right)$. (ii)(c) $\theta = 5.51$.

DR ESLAM AHMED

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7 marks

WMA12/01 MAY/JUNE (R) 2024

Question 7

Past-paper question

WMA12-01_2024_MayJuneR_Q07

7. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 \leq x < 2\pi$, the equation

$$3 \sin x \tan x = 11 + \cos x$$

giving the answers in radians to 3 decimal places.

(5)

- (ii) Given that

- $0 < \theta < 90^\circ$
- $\cos \theta = \frac{1}{3}$

find, in simplest form, the exact value of $\tan \theta$

(2)

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WORKED SOLUTION

Question 7

WMA12/01 May/June (R) 2024

WMA12-01_2024_MayJuneR_Q07

1

Rewrite tan x $3 \sin x \tan x = 11 + \cos x$ becomes $\frac{3 \sin^2 x}{\cos x} = 11 + \cos x$.

2

Multiply by cos x $3 \sin^2 x = 11 \cos x + \cos^2 x$.

3

Use sine squared $3(1 - \cos^2 x) = 11 \cos x + \cos^2 x$, so $4 \cos^2 x + 11 \cos x - 3 = 0$.

4

Solve for cosine $(4 \cos x - 1)(\cos x + 3) = 0$, so $\cos x = \frac{1}{4}$ or $\cos x = -3$.

5

Reject impossible value $\cos x = -3$ is impossible, so $\cos x = \frac{1}{4}$.

6

Find x valuesFor $0 \leq x < 2\pi$, this gives $x = 1.318 \dots$ or $x = 4.965 \dots$, so $x = 1.318, 4.965$ radians.

7

Use the right triangle identityGiven $0 < \theta < 90^\circ$ and $\cos \theta = \frac{1}{3}$, $\sin \theta = \sqrt{1 - \frac{1}{9}} = \frac{2\sqrt{2}}{3}$.

8

Find tan theta $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{2\sqrt{2}/3}{1/3} = 2\sqrt{2}$.**FINAL ANSWER**

WMA12-01_2024_MayJuneR_Q07

(i) $x = 1.318, 4.965$. (ii) $2\sqrt{2}$.

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8 marks

WMA12/01 MAY/JUNE 2025

Question 8

Past-paper question

WMA12-01_2025_MayJune_Q08

8:

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

- (i) Solve, for $0 \leq x < \pi$, the equation

$$3 \tan \left(2x + \frac{\pi}{5} \right) = \sqrt{3}$$

giving the answers in radians in the form $k\pi$, where k is a rational constant to be found.

(3)

- (ii) Solve, for $0 \leq \theta < 360^\circ$, the equation

$$5 \sin \theta \tan \theta = \cos \theta + 4$$

giving your answers, in degrees, to one decimal place.

(5)

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WORKED SOLUTION

Question 8

WMA12/01 May/June 2025

WMA12-01_2025_MayJune_Q08

1

Solve the tangent equation

$$3 \tan \left(2x + \frac{\pi}{5} \right) = \sqrt{3} \text{ gives } \tan \left(2x + \frac{\pi}{5} \right) = \frac{1}{\sqrt{3}}.$$

2

Write the general angle

$$\text{Therefore } 2x + \frac{\pi}{5} = \frac{\pi}{6} + n\pi.$$

3

Find x

$$2x = -\frac{\pi}{30} + n\pi, \text{ so } x = -\frac{\pi}{60} + \frac{n\pi}{2}.$$

4

Use the interval

$$\text{For } 0 \leq x < \pi, \text{ the possible values are } x = \frac{29\pi}{60} \text{ and } x = \frac{59\pi}{60}.$$

5

Rewrite the second equation

$$\text{Using } \tan \theta = \frac{\sin \theta}{\cos \theta}, 5 \sin \theta \tan \theta = \cos \theta + 4 \text{ becomes } \frac{5 \sin^2 \theta}{\cos \theta} = \cos \theta + 4.$$

6

Form an equation in cos theta

$$\text{Multiplying by } \cos \theta \text{ gives } 5 \sin^2 \theta = \cos^2 \theta + 4 \cos \theta. \text{ Using } \sin^2 \theta = 1 - \cos^2 \theta \text{ gives } 6 \cos^2 \theta + 4 \cos \theta - 5 = 0.$$

7

Solve for cos theta

$$\cos \theta = \frac{-2 \pm \sqrt{34}}{6}. \text{ The negative option is less than } -1, \text{ so reject it.}$$

8

Find theta

$$\cos \theta = \frac{-2 + \sqrt{34}}{6} \text{ gives } \theta = 50.320 \dots^\circ \text{ or } 309.679 \dots^\circ.$$

9

Round the answers

$$\text{To one decimal place, } \theta = 50.3^\circ \text{ or } 309.7^\circ.$$

FINAL ANSWER

$$(i) x = \frac{29\pi}{60}, \frac{59\pi}{60}. \quad (ii) \theta = 50.3^\circ, 309.7^\circ.$$

WMA12-01_2025_MayJune_Q08

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9 marks

WMA12/01 OCTOBER 2025

Question 5

Past-paper question

WMA12-01_2025_Oct_Q05

5. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 < \theta \leq 360^\circ$, the equation

$$4 \tan \theta + 5 \sin \theta = 0$$

giving any non-exact answers to one decimal place.

(5)

- (ii) Solve, for $0 < x < \pi$, the equation

$$\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{5}{\cos x}$$

giving the answers, in radians, to 3 significant figures.

(4)

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WORKED SOLUTION

Question 5

WMA12/01 October 2025

WMA12-01_2025_Oct_Q05

1

Rewrite the first equation $4 \tan \theta + 5 \sin \theta = 0$ gives $4 \frac{\sin \theta}{\cos \theta} + 5 \sin \theta = 0$.

2

Factor sin theta $\sin \theta \left(\frac{4}{\cos \theta} + 5 \right) = 0$.

3

Use sin theta equals zeroIn $0 < \theta \leq 360^\circ$, $\sin \theta = 0$ gives $\theta = 180^\circ$ or 360° .

4

Use the other factor $\frac{4}{\cos \theta} + 5 = 0$ gives $\cos \theta = -\frac{4}{5}$.

5

Find the remaining anglesFor $0 < \theta \leq 360^\circ$, $\cos \theta = -\frac{4}{5}$ gives $\theta = 143.130 \dots^\circ$ or $216.869 \dots^\circ$.

6

Round part iThe non-exact angles are 143.1° and 216.9° to one decimal place.

7

Clear the fractions in part ii $\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{5}{\cos x}$. Multiplying by $\sin x \cos x$ gives $\sin^2 x + \cos^2 x = 5 \sin x$.

8

Use the identity $\sin^2 x + \cos^2 x = 1$, so $1 = 5 \sin x$ and $\sin x = \frac{1}{5}$.

9

Solve in the intervalFor $0 < x < \pi$, $x = \sin^{-1} \left(\frac{1}{5} \right)$ or $x = \pi - \sin^{-1} \left(\frac{1}{5} \right)$.

10

Round part iiThus $x = 0.201357 \dots$ or $2.94023 \dots$, so $x = 0.201$ or 2.94 to 3 significant figures.

FINAL ANSWER

WMA12-01_2025_Oct_Q05

(i) $\theta = 143.1^\circ, 180^\circ, 216.9^\circ, 360^\circ$. (ii) $x = 0.201, 2.94$.

DR ESLAM AHMED

TOPIC 10 | FAMILY 02

Trig Identities

WHAT THIS FAMILY ASKS

Recognise the trig identities demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

2 classified questions

15 total marks

T10

10_IDENT

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CHAPTER

Trigonometric Equations

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Trig identities

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7 marks

WMA12/01 JANUARY 2020

Question 7

Past-paper question

WMA12-01_2020_Jan_Q07

7. (a) Show that the equation

$$8 \tan \theta = 3 \cos \theta$$

may be rewritten in the form

$$3 \sin^2 \theta + 8 \sin \theta - 3 = 0 \quad (3)$$

- (b) Hence solve, for $0 \leq x \leq 90^\circ$, the equation

$$8 \tan 2x = 3 \cos 2x$$

giving your answers to 2 decimal places. (4)

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WORKED SOLUTION

Question 7

WMA12/01 January 2020

WMA12-01_2020_Jan_Q07

1

Rewrite tan theta $8 \tan \theta = 3 \cos \theta$ becomes $8 \frac{\sin \theta}{\cos \theta} = 3 \cos \theta$.

2

Clear the denominatorMultiplying by $\cos \theta$ gives $8 \sin \theta = 3 \cos^2 \theta$.

3

Use the identitySince $\cos^2 \theta = 1 - \sin^2 \theta$, $8 \sin \theta = 3(1 - \sin^2 \theta)$, so $3 \sin^2 \theta + 8 \sin \theta - 3 = 0$.

4

Apply the result to $2x$ For $8 \tan 2x = 3 \cos 2x$, we get $3 \sin^2(2x) + 8 \sin(2x) - 3 = 0$.

5

Solve for $\sin 2x$ $(3 \sin 2x - 1)(\sin 2x + 3) = 0$. Since $\sin 2x$ cannot be -3 , $\sin 2x = \frac{1}{3}$.

6

Find x For $0 \leq x \leq 90^\circ$, $0 \leq 2x \leq 180^\circ$. Thus $2x = 19.471 \dots^\circ$ or $160.529 \dots^\circ$, so $x = 9.74^\circ$ or 80.26° .**FINAL ANSWER**

WMA12-01_2020_Jan_Q07

(a) **Proven.** (b) $x = 9.74^\circ, 80.26^\circ$.

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WMA12/01 JANUARY 2025

Question 10

Past-paper question

WMA12-01_2025_Jan_Q10

10. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$\cos \theta \left(3 \tan \theta + \frac{2}{\tan \theta} \right) \equiv \sin \theta + \frac{2}{\sin \theta} \quad \theta \neq \frac{n\pi}{2} \quad (4)$$

(b) Hence solve, for $0 < x < 2\pi$, the equation

$$\cos x \left(3 \tan x + \frac{2}{\tan x} \right) = 4 \sin x - 5$$

giving your answers to 3 significant figures.

(4)

T10 Trigonometric Equations 10_IDENT Trig identities

10_IDENT_CORE core family
move

WORKED SOLUTION

Question 10

WMA12/01 January 2025

WMA12-01_2025_Jan_Q10

1

Expand the left side

$$\cos \theta \left(3 \tan \theta + \frac{2}{\tan \theta} \right) = 3 \sin \theta + \frac{2 \cos \theta}{\tan \theta}.$$

2

Rewrite the second term

$$\text{Since } \tan \theta = \frac{\sin \theta}{\cos \theta}, \frac{2 \cos \theta}{\tan \theta} = \frac{2 \cos^2 \theta}{\sin \theta}.$$

3

Use the identity

$$3 \sin \theta + \frac{2 \cos^2 \theta}{\sin \theta} = 3 \sin \theta + \frac{2(1 - \sin^2 \theta)}{\sin \theta}.$$

4

Simplify

$$3 \sin \theta + \frac{2}{\sin \theta} - 2 \sin \theta = \sin \theta + \frac{2}{\sin \theta}, \text{ as required.}$$

5

Use part a

$$\text{The equation becomes } \sin x + \frac{2}{\sin x} = 4 \sin x - 5.$$

6

Form a quadratic in $\sin x$

$$\text{Multiplying by } \sin x \text{ gives } \sin^2 x + 2 = 4 \sin^2 x - 5 \sin x, \text{ so } 3 \sin^2 x - 5 \sin x - 2 = 0.$$

7

Solve for $\sin x$

$$(3 \sin x + 1)(\sin x - 2) = 0. \text{ Since } \sin x = 2 \text{ is impossible, } \sin x = -\frac{1}{3}.$$

8

Find x in the interval

$$\text{For } 0 < x < 2\pi, \sin x = -\frac{1}{3} \text{ gives } x = \pi + \sin^{-1}\left(\frac{1}{3}\right) \text{ or } x = 2\pi - \sin^{-1}\left(\frac{1}{3}\right).$$

9

Round the answers

$$\text{Thus } x = 3.481 \dots \text{ or } x = 5.943 \dots, \text{ so to 3 significant figures } x = 3.48 \text{ or } x = 5.94.$$

FINAL ANSWER

WMA12-01_2025_Jan_Q10

(a) Proven. (b) $x = 3.48, 5.94$.

TOPIC 10 | FAMILY 03

Periodicity

WHAT THIS FAMILY ASKS

Recognise the periodicity demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

1 classified questions

10 total marks

T10

10_PERIOD

10_PERIOD_CORE

CHAPTER

Trigonometric Equations

FAMILY

Periodicity

VARIANT

core family move

10 marks

WMA12/01 OCTOBER 2021

Question 10

Past-paper question

WMA12-01_2021_Oct_Q10

10. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

$$\tan^2\left(2x + \frac{\pi}{4}\right) = 3 \quad (5)$$

- (ii) Solve, for $0 < \theta < 360^\circ$

$$(2 \sin \theta - \cos \theta)^2 = 1$$

giving your answers, as appropriate, to one decimal place.

(5)

(Total 10 marks)

T10 Trigonometric Equations 10_PERIOD Periodicity

10_PERIOD_CORE core
family move

WORKED SOLUTION

Question 10

WMA12/01 October 2021

WMA12-01_2021_Oct_Q10

1

Take the square rootLet $u = 2x + \frac{\pi}{4}$. From $\tan^2 u = 3$, $\tan u = \pm\sqrt{3}$.

2

Write the angle familiesThus $u = \frac{\pi}{3} + n\pi$ or $u = -\frac{\pi}{3} + n\pi$, where n is an integer.

3

Return to x So $x = \frac{\pi}{24} + \frac{n\pi}{2}$ or $x = -\frac{7\pi}{24} + \frac{n\pi}{2}$.

4

Use the intervalFor $-\frac{\pi}{2} < x < \frac{\pi}{2}$, the solutions are $x = -\frac{11\pi}{24}, -\frac{7\pi}{24}, \frac{\pi}{24}, \frac{5\pi}{24}$.

5

Expand the squared equation $(2 \sin \theta - \cos \theta)^2 = 1$ gives $4 \sin^2 \theta - 4 \sin \theta \cos \theta + \cos^2 \theta = 1$.

6

Use the identitySince $1 = \sin^2 \theta + \cos^2 \theta$, this simplifies to $3 \sin^2 \theta - 4 \sin \theta \cos \theta = 0$.

7

Factor $\sin \theta (3 \sin \theta - 4 \cos \theta) = 0$.

8

Solve the two cases $\sin \theta = 0$ gives $\theta = 180^\circ$ in the interval. Also $3 \sin \theta = 4 \cos \theta$ gives $\tan \theta = \frac{4}{3}$.

9

State thetaFor $0 < \theta < 360^\circ$, $\tan \theta = \frac{4}{3}$ gives $\theta = 53.1^\circ$ and 233.1° . Therefore $\theta = 53.1^\circ, 180^\circ, 233.1^\circ$.**FINAL ANSWER**

WMA12-01_2021_Oct_Q10

(i) $x = -\frac{11\pi}{24}, -\frac{7\pi}{24}, \frac{\pi}{24}, \frac{5\pi}{24}$. (ii) $\theta = 53.1^\circ, 180^\circ, 233.1^\circ$.

TOPIC 10 | FAMILY 04

Transformed Trig

WHAT THIS FAMILY ASKS

Recognise the transformed trig demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

V02 Expand the interval for the inside expression before finding angles.

8 classified questions

69 total marks

T10

10_TRANS

10_TRANS_CORE

CHAPTER

Trigonometric Equations

FAMILY

Transformed trig

VARIANT

core family move

10 marks

WMA12/01 MAY/JUNE 2021

Question 8

Past-paper question

WMA12-01_2021_MayJune_Q08

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 < \theta < 360^\circ$, the equation

$$3 \sin(\theta + 30^\circ) = 7 \cos(\theta + 30^\circ)$$

giving your answers to one decimal place.

(4)

- (ii) (a) Show that the equation

$$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$$

can be written in the form

$$\sin x(a \cos^2 x + b \cos x + c) = 0$$

where a , b and c are constants to be found.

- (b) Hence solve for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ the equation

$$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$$

(6)

T10 Trigonometric Equations

10_TRANS

Transformed trig

10_TRANS_CORE core family
move

WORKED SOLUTION

Question 8

WMA12/01 May/June 2021

WMA12-01_2021_MayJune_Q08

1

Solve for theta plus thirtyLet $u = \theta + 30^\circ$. Then $3 \sin u = 7 \cos u$, so $\tan u = \frac{7}{3}$.

2

Find u in the intervalSince $0 < \theta < 360^\circ$, $30^\circ < u < 390^\circ$. Thus $u = 66.801 \dots^\circ$ or $246.801 \dots^\circ$.

3

Return to theta $\theta = u - 30^\circ$, giving $\theta = 36.8^\circ$ and 216.8° to one decimal place.

4

Rearrange the second equation $3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$ gives $3 \sin^3 x - 5 \sin x + 7 \sin x \cos x = 0$.

5

Factor out sine $\sin x(3 \sin^2 x - 5 + 7 \cos x) = 0$.

6

Use sin squaredSince $\sin^2 x = 1 - \cos^2 x$, this becomes $\sin x(-3 \cos^2 x + 7 \cos x - 2) = 0$, or $\sin x(3 \cos^2 x - 7 \cos x + 2) = 0$.

7

Solve in the intervalFor $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$, $\sin x = 0$ gives $x = 0$. Also $3 \cos^2 x - 7 \cos x + 2 = 0$ gives $(3 \cos x - 1)(\cos x - 2) = 0$.

8

State the solutions $\cos x = 2$ is impossible, so $\cos x = \frac{1}{3}$. The solutions are $x = -\cos^{-1}\left(\frac{1}{3}\right)$, 0, and $\cos^{-1}\left(\frac{1}{3}\right)$.**FINAL ANSWER**

WMA12-01_2021_MayJune_Q08

(i) $\theta = 36.8^\circ, 216.8^\circ$. (ii) (a) $a = 3, b = -7, c = 2$. (ii) (b) $x = -\cos^{-1}\left(\frac{1}{3}\right), 0, \cos^{-1}\left(\frac{1}{3}\right)$.

T10

10_TRANS

10_TRANS_CORE

CHAPTER

Trigonometric Equations

FAMILY

Transformed trig

VARIANT

core family move

9 marks

WMA12/01 JANUARY 2023

Question 8

Past-paper question

WMA12-01_2023_Jan_Q08

8. In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

- (i) Solve, for $-\frac{\pi}{2} < x < \pi$, the equation

$$5 \sin(3x + 0.1) + 2 = 0$$

giving your answers, **in radians**, to 2 decimal places.

(4)

- (ii) Solve, for $0 < \theta < 360^\circ$, the equation

$$2 \tan \theta \sin \theta = 5 + \cos \theta$$

giving your answers, **in degrees**, to one decimal place.

(5)

T10 Trigonometric Equations

10_TRANS

Transformed trig

10_TRANS_CORE

core family

move

WORKED SOLUTION

Question 8

WMA12/01 January 2023

WMA12-01_2023_Jan_Q08

1

Isolate the sine term $5 \sin(3x + 0.1) + 2 = 0$ gives $\sin(3x + 0.1) = -0.4$.

2

Find the angle intervalSince $-\frac{\pi}{2} < x < \pi$, multiplying by 3 and adding 0.1 gives $-\frac{3\pi}{2} + 0.1 < 3x + 0.1 < 3\pi + 0.1$.

3

Solve for the inner angleIn this interval, $3x + 0.1 = -2.7301 \dots, -0.4115 \dots, 3.5531 \dots, 5.8717 \dots$

4

Find x valuesSubtract 0.1 and divide by 3: $x = -0.9433 \dots, -0.1705 \dots, 1.1510 \dots, 1.9239 \dots$

5

Round the radian answersTo 2 decimal places, $x = -0.94, -0.17, 1.15, 1.92$.

6

Rewrite the second equation $2 \tan \theta \sin \theta = 5 + \cos \theta$. Using $\tan \theta = \frac{\sin \theta}{\cos \theta}$ gives $\frac{2 \sin^2 \theta}{\cos \theta} = 5 + \cos \theta$.

7

Use sine squaredMultiplying by $\cos \theta$ gives $2 \sin^2 \theta = 5 \cos \theta + \cos^2 \theta$. Substitute $\sin^2 \theta = 1 - \cos^2 \theta$.

8

Solve for cos theta $2(1 - \cos^2 \theta) = 5 \cos \theta + \cos^2 \theta$, so $3 \cos^2 \theta + 5 \cos \theta - 2 = 0$.

9

Factor and reject impossible value $(3 \cos \theta - 1)(\cos \theta + 2) = 0$. Since $\cos \theta = -2$ is impossible, $\cos \theta = \frac{1}{3}$.

10

Find thetaFor $0 < \theta < 360^\circ$, $\cos \theta = \frac{1}{3}$ gives $\theta = 70.5^\circ$ or 289.5° to one decimal place.

FINAL ANSWER

WMA12-01_2023_Jan_Q08

(i) $x = -0.94, -0.17, 1.15, 1.92$. (ii) $\theta = 70.5^\circ, 289.5^\circ$.

DR ESLAM AHMED

T10

10_TRANS

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CHAPTER

Trigonometric Equations

FAMILY

Transformed trig

VARIANT

core family move

12 marks

WMA12/01 MAY/JUNE 2024

Question 8

Past-paper question

WMA12-01_2024_MayJune_Q08

8. **In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

- (i) Solve, for $0 < x \leq \pi$, the equation

$$5 \sin x \tan x + 13 = \cos x$$

giving your answer in radians to 3 significant figures.

(5)

- (ii) The temperature inside a greenhouse is monitored on one particular day.

The temperature, $H^\circ\text{C}$, inside the greenhouse, t hours after midnight, is modelled by the equation

$$H = 10 + 12 \sin(kt + 18)^\circ \quad 0 \leq t < 24$$

where k is a constant.

Use the equation of the model to answer parts (a) to (c).

Given that

- the temperature inside the greenhouse was 20°C at 6 am
- $0 < k < 20$

- (a) find all possible values for k , giving each answer to 2 decimal places.

(4)

Given further that $0 < k < 10$

- (b) find the maximum temperature inside the greenhouse,
- (c) find the time of day at which this maximum temperature occurs.

(1)

Give your answer to the nearest minute.

(2)

T10 Trigonometric Equations

10_TRANS

Transformed trig

10_TRANS_CORE core family
move

WORKED SOLUTION

Question 8

WMA12/01 May/June 2024

WMA12-01_2024_MayJune_Q08

1

Rewrite the trig equation $5 \sin x \tan x + 13 = \cos x$ becomes $\frac{5 \sin^2 x}{\cos x} + 13 = \cos x$.

2

Multiply by $\cos x$ $5 \sin^2 x + 13 \cos x = \cos^2 x$.

3

Use sine squared $5(1 - \cos^2 x) + 13 \cos x = \cos^2 x$, so $6 \cos^2 x - 13 \cos x - 5 = 0$.

4

Solve for $\cos x$ $(2 \cos x - 5)(3 \cos x + 1) = 0$, so $\cos x = \frac{5}{2}$ or $\cos x = -\frac{1}{3}$.

5

Choose the valid root $\cos x = \frac{5}{2}$ is impossible, so $\cos x = -\frac{1}{3}$. For $0 < x \leq \pi$, $x = 1.9106 \dots$, so $x = 1.91$ radians to 3 significant figures.

6

Use the temperature at 6 amAt $t = 6$, $20 = 10 + 12 \sin(6k + 18)^\circ$, so $\sin(6k + 18)^\circ = \frac{5}{6}$.

7

Solve for k $\sin^{-1}\left(\frac{5}{6}\right) = 56.442 \dots^\circ$. Since $0 < k < 20$, $18 < 6k + 18 < 138$, so $6k + 18 = 56.442 \dots$ or $123.557 \dots$.

8

Find possible k valuesThis gives $k = 6.407 \dots$ or $k = 17.592 \dots$, so $k = 6.41$ or 17.59 to 2 decimal places.

9

Use the tighter conditionGiven further that $0 < k < 10$, use $k = 6.407 \dots$.

10

Find the maximum temperatureThe maximum value of $\sin$ is 1, so the maximum temperature is $10 + 12 = 22^\circ\text{C}$.

11

Find the time of maximum

The maximum occurs when $kt + 18 = 90$, so $t = \frac{72}{6.407...} = 11.2375 \dots$ hours after midnight.

12

Convert to time of day

0.2375 ... hours is 14.25 ... minutes, so the maximum occurs at about 11:14 am.

FINAL ANSWER

WMA12-01_2024_MayJune_Q08

(i) $x = 1.91$ **radians.** (ii) (a) $k = 6.41$ **or** 17.59. (ii) (b) 22° **C.** (ii) (c) 11:14 **am.**

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CHAPTER

Trigonometric Equations

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6 marks

WMA12/01 MAY/JUNE (R) 2024

Question 5

Past-paper question

WMA12-01_2024_MayJuneR_Q05

5.

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

The depth of water, D metres, in a harbour on a particular day is given by the equation

$$D = 8 + 5 \sin\left(\frac{\pi t}{6} + 3\right) \quad 0 \leq t < 24$$

where t is the number of hours after **midnight**.

- (a) Show that the depth of water in the harbour at 2 am is just over 4 metres. (1)
- (b) Find, to the nearest minute, the first time after **midday** when the depth of water in the harbour is exactly 6 metres. (5)

T10 Trigonometric Equations

10_TRANS

Transformed trig

10_TRANS_CORE core family
move

WORKED SOLUTION

Question 5

WMA12/01 May/June (R) 2024

WMA12-01_2024_MayJuneR_Q05

1

Substitute t equals two

$$D = 8 + 5 \sin \left(\frac{\pi(2)}{6} + 3 \right) = 8 + 5 \sin \left(\frac{\pi}{3} + 3 \right).$$

2

Evaluate

$D = 4.064 \dots$, so the depth at 2 am is just over 4 metres.

3

Set depth equal to six

$$6 = 8 + 5 \sin \left(\frac{\pi t}{6} + 3 \right), \text{ so } \sin \left(\frac{\pi t}{6} + 3 \right) = -\frac{2}{5}.$$

4

Use the time interval

After midday means $t > 12$. Then $\frac{\pi t}{6} + 3$ is greater than $2\pi + 3$.

5

Choose the first suitable angle

The first angle after $2\pi + 3$ with sine $-\frac{2}{5}$ is $3\pi + \sin^{-1} \left(\frac{2}{5} \right)$.

6

Solve for t

$$\frac{\pi t}{6} + 3 = 3\pi + \sin^{-1} \left(\frac{2}{5} \right), \text{ so } t = 13.056 \dots \text{ hours after midnight.}$$

7

Convert to clock time

$0.056 \dots$ hours is about 3.38 minutes, so the time to the nearest minute is 13:03, or 1:03 pm.

FINAL ANSWER

WMA12-01_2024_MayJuneR_Q05

(a) $D(2) = 4.06 \dots$ m, just over 4 m. (b) 13:03, or 1:03 pm.

T10

10_TRANS

10_TRANS_CORE

CHAPTER

Trigonometric Equations

FAMILY

Transformed trig

VARIANT

core family move

7 marks

WMA12/01 OCTOBER 2024

Question 9

Past-paper question

WMA12-01_2024_Oct_Q09

9.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that the equation

$$2 \tan \theta = 3 \cos \theta$$

can be written as

$$3 \sin^2 \theta + 2 \sin \theta - 3 = 0$$

(3)

(b) Hence solve, for $-\pi < x < \pi$, the equation

$$2 \tan \left(2x + \frac{\pi}{3} \right) = 3 \cos \left(2x + \frac{\pi}{3} \right)$$

giving your answers to 3 significant figures.

(4)

T10 Trigonometric Equations

10_TRANS

Transformed trig

10_TRANS_CORE core family
move

WORKED SOLUTION

Question 9

WMA12/01 October 2024

WMA12-01_2024_Oct_Q09

1

Write tan in terms of sin and cos $2 \tan \theta = 3 \cos \theta$ gives $2 \frac{\sin \theta}{\cos \theta} = 3 \cos \theta$.

2

Multiply by cos thetaSince $\tan \theta$ is defined, $\cos \theta \neq 0$. Hence $2 \sin \theta = 3 \cos^2 \theta$.

3

Use the identity $2 \sin \theta = 3(1 - \sin^2 \theta)$, so $3 \sin^2 \theta + 2 \sin \theta - 3 = 0$.

4

Substitute the new angleLet $\theta = 2x + \frac{\pi}{3}$. Then $3 \sin^2 \theta + 2 \sin \theta - 3 = 0$.

5

Solve for sin theta $\sin \theta = \frac{-2 \pm \sqrt{40}}{6} = \frac{-1 \pm \sqrt{10}}{3}$. The root $\frac{-1 - \sqrt{10}}{3}$ is less than -1 , so reject it.

6

Find the angle rangeSince $-\pi < x < \pi$, $-\frac{5\pi}{3} < \theta < \frac{7\pi}{3}$.

7

Find the theta valuesFor $\sin \theta = \frac{-1 + \sqrt{10}}{3}$ in this range, $\theta \approx -3.94649, 0.804897, 2.33670$, and 7.08808 .

8

Convert back to xUsing $x = \frac{\theta - \pi/3}{2}$ gives $x \approx -2.49684, -0.121150, 0.644749$, and 3.02044 .

9

Round the answersTo 3 significant figures, $x = -2.50, -0.121, 0.645$, and 3.02 .**FINAL ANSWER**

WMA12-01_2024_Oct_Q09

(a) **Proven.** (b) $x = -2.50, -0.121, 0.645, 3.02$.

T10

10_TRANS

10_TRANS_HV01

CHAPTER

Trigonometric Equations

FAMILY

Transformed trig

VARIANT

Expand the interval for the
inside expression before
finding angles.

8 marks

WMA12/01 OCTOBER 2022

Question 5

Past-paper question

WMA12-01_2022_Oct_Q05

5.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that the equation

$$(3 \cos \theta - \tan \theta) \cos \theta = 2$$

can be written as

$$3 \sin^2 \theta + \sin \theta - 1 = 0$$

(3)

(b) Hence solve for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

$$(3 \cos 2x - \tan 2x) \cos 2x = 2$$

(5)

T10 Trigonometric Equations 10_TRANS Transformed trig

10_TRANS_HV01 Expand the interval for the inside expression before finding angles.

WORKED SOLUTION

Question 5

WMA12/01 October 2022

WMA12-01_2022_Oct_Q05

1

Rewrite the left side $(3 \cos \theta - \tan \theta) \cos \theta = 3 \cos^2 \theta - \sin \theta$ because $\tan \theta \cos \theta = \sin \theta$.

2

Move everything to one sideThe equation becomes $3 \cos^2 \theta - \sin \theta = 2$.

3

Use cos squaredSince $\cos^2 \theta = 1 - \sin^2 \theta$, $3(1 - \sin^2 \theta) - \sin \theta = 2$.

4

Show the required formExpanding and rearranging gives $3 \sin^2 \theta + \sin \theta - 1 = 0$.

5

Apply the result to 2xFor $(3 \cos 2x - \tan 2x) \cos 2x = 2$, take $\theta = 2x$. Then $3 \sin^2(2x) + \sin(2x) - 1 = 0$.

6

Solve the quadraticLet $u = \sin(2x)$. Then $3u^2 + u - 1 = 0$, so $u = \frac{-1 \pm \sqrt{13}}{6}$.

7

Use the intervalSince $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$, we have $-\pi \leq 2x \leq \pi$. Solving $\sin(2x) = \frac{-1 + \sqrt{13}}{6}$ and $\sin(2x) = \frac{-1 - \sqrt{13}}{6}$ in this interval gives $2x = -2.2665 \dots, -0.8751 \dots, 0.4492 \dots, 2.6924 \dots$

8

Divide by twoTherefore $x = -1.133 \dots, -0.4375 \dots, 0.2246 \dots, 1.346 \dots$, giving $x = -1.13, -0.438, 0.225, 1.35$ radians.**FINAL ANSWER**

WMA12-01_2022_Oct_Q05

(a) Proven. (b) $x = -1.13, -0.438, 0.225, 1.35$ radians.

T10

10_TRANS

10_TRANS_HV01

CHAPTER

Trigonometric Equations

FAMILY

Transformed trig

VARIANT

Expand the interval for the
inside expression before
finding angles.

7 marks

WMA12/01 MAY/JUNE 2023

Question 9

Past-paper question

WMA12-01_2023_MayJune_Q09

9.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

(a) Show that

$$3 \cos \theta (\tan \theta \sin \theta + 3) = 11 - 5 \cos \theta$$

may be written as

$$3 \cos^2 \theta - 14 \cos \theta + 8 = 0 \quad (3)$$

(b) Hence solve, for $0 < x < 360^\circ$

$$3 \cos 2x (\tan 2x \sin 2x + 3) = 11 - 5 \cos 2x$$

giving your answers to one decimal place.

(4)

T10 Trigonometric Equations 10_TRANS Transformed trig

10_TRANS_HV01 Expand the interval for the inside expression before finding angles.

WORKED SOLUTION

Question 9

WMA12/01 May/June 2023

WMA12-01_2023_MayJune_Q09

1

Rewrite $\tan \theta \sin \theta$

$$\tan \theta \sin \theta = \frac{\sin^2 \theta}{\cos \theta}.$$

2

Simplify the left side

$$3 \cos \theta (\tan \theta \sin \theta + 3) = 3 \sin^2 \theta + 9 \cos \theta.$$

3

Use sine squared

$$\text{The equation becomes } 3(1 - \cos^2 \theta) + 9 \cos \theta = 11 - 5 \cos \theta.$$

4

Rearrange

$$3 - 3 \cos^2 \theta + 9 \cos \theta = 11 - 5 \cos \theta, \text{ so } 3 \cos^2 \theta - 14 \cos \theta + 8 = 0.$$

5

Apply the result to $2x$

$$\text{For the equation in part (b), take } \theta = 2x. \text{ Then } 3 \cos^2(2x) - 14 \cos(2x) + 8 = 0.$$

6

Solve the quadratic

$$\text{Let } u = \cos(2x). \text{ Then } 3u^2 - 14u + 8 = 0, \text{ so } (3u - 2)(u - 4) = 0.$$

7

Reject the impossible value

$$u = 4 \text{ is impossible for cosine, so } \cos(2x) = \frac{2}{3}.$$

8

Use the interval

$$\text{Since } 0 < x < 360^\circ, \text{ we have } 0 < 2x < 720^\circ.$$

9

Find x

In this interval, $2x = 48.189 \dots^\circ, 311.810 \dots^\circ, 408.189 \dots^\circ, 671.810 \dots^\circ$. Dividing by 2 gives $x = 24.1^\circ, 155.9^\circ, 204.1^\circ, 335.9^\circ$ to one decimal place.

FINAL ANSWER

WMA12-01_2023_MayJune_Q09

(a) Proven. (b) $x = 24.1^\circ, 155.9^\circ, 204.1^\circ, 335.9^\circ$.

T10

10_TRANS

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CHAPTER

Trigonometric Equations

FAMILY

Transformed trig

VARIANT

Expand the interval for the
inside expression before
finding angles.

10 marks

WMA12/01 JANUARY 2026

Question 8

Past-paper question

WMA12-01_2026_Jan_Q08

8.

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

- (i) Solve, for $0 \leq \theta < 180$

$$4 \tan^2(2\theta - 30)^\circ + 1 = 49$$

giving the answers to one decimal place.

(5)

- (ii) Solve, for $0 \leq x < 2\pi$

$$2 \tan x \sin x + 3 = 0$$

giving the answers, in radians, in the form $k\pi$ where k is a rational constant.

(5)

T10 Trigonometric Equations 10_TRANS Transformed trig

10_TRANS_HV01 Expand the interval for the inside expression before finding angles.

WORKED SOLUTION

Question 8

WMA12/01 January 2026

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1

Make the tangent expression the subject $4 \tan^2(2\theta - 30^\circ) + 1 = 49$ gives $\tan^2(2\theta - 30^\circ) = 12$.

2

Find the possible angle valuesLet $\alpha = 2\theta - 30^\circ$. Since $0 \leq \theta < 180^\circ$, $-30^\circ \leq \alpha < 330^\circ$. The solutions of $\tan \alpha = \pm\sqrt{12}$ in this range are approximately $73.9^\circ, 106.1^\circ, 253.9^\circ, 286.1^\circ$.

3

Return to theta $\theta = \frac{\alpha + 30^\circ}{2}$, giving $\theta = 51.9^\circ, 68.1^\circ, 141.9^\circ, 158.1^\circ$ to one decimal place.

4

Rewrite $\tan x$ $2 \tan x \sin x + 3 = 0$ becomes $2 \left(\frac{\sin x}{\cos x} \right) \sin x + 3 = 0$.

5

Form a quadratic in $\cos x$ Multiplying by $\cos x$ gives $2 \sin^2 x + 3 \cos x = 0$. Using $\sin^2 x = 1 - \cos^2 x$, $2(1 - \cos^2 x) + 3 \cos x = 0$, so $2 \cos^2 x - 3 \cos x - 2 = 0$.

6

Solve for x $(2 \cos x + 1)(\cos x - 2) = 0$. Since $\cos x = 2$ is impossible, $\cos x = -\frac{1}{2}$. In $0 \leq x < 2\pi$, $x = \frac{2\pi}{3}$ or $x = \frac{4\pi}{3}$.**FINAL ANSWER**

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(i) $\theta = 51.9^\circ, 68.1^\circ, 141.9^\circ, 158.1^\circ$. (ii) $x = \frac{2\pi}{3}, \frac{4\pi}{3}$.